Yu-Han Lee (Tyler)

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SUMMARY

A passionate technology enthusiast with experience in developing web applications and machine learning models. Skilled in deep learning and cross-functional collaboration, committed to driving impactful projects. Strong foundation in data-driven methodologies, ready to contribute to transformative initiatives and positive outcomes.

SKILLS & LANGUAGES

Skills:

- **Programming Languages**: Python, C++, Java
- **Tools/Frameworks**: TensorFlow, OpenCV, NumPy, pandas, Git, Docker, Keras
- **Data Analysis**: Jupyter Notebooks, Matplotlib

Languages: Mandarin Chinese (native); English (fluent)

EDUCATION

King's College London — **London, United Kingdom**
Master's Degree in Advanced Computing — **Sep 2024 – present**

- **Grade:** Pending

Chung Yuan Christian University — **Taoyuan, Taiwan**
Bachelor in Information and Computer Engineering — **Sep 2020 – Jan 2024**
Department of Commercial Design (Product Design Division) — **Sep 2019 – Sep 2020**
(Transferred to the Department of Information and Computer Engineering in September 2020)

- **GPA:** 3.93/4.0
- **Overall:** 88.79%
- Received the Dean's List Award for ranking 2nd out of 56 (top 3%) in class in the spring semester 2020.

EMPLOYMENT

Chang Ching Yu Memorial Library, Chung Yuan Christian University — **Taoyuan, Taiwan**
Library IT Intern — **Feb 2022 – Feb 2024**

- Enhanced a Python-based thesis checker, reducing processing time by 25% and improving accuracy by 30%.
- Provided technical support to international students, facilitating communication during the review process.
- Led cross-functional teams to resolve workflow inefficiencies, improving system reliability by 20% and reducing incident resolution time by 15%.

Taiwan Mobile Co., Ltd. — **Taipei, Taiwan**
Engineer Intern, Web Technology Division, Customer Service System Dept. — **Jul 2023 – Aug 2023**

- Optimized website performance, improving SEO rankings by 20%, increasing traffic by 15%.
- Formulated Java-based image conversion tools, boosting processing speed and accuracy.
- Devised methods for activating mobile cameras using HTML5, W3C MediaStream API, Vue.js, and jQuery.
- Designed a YOLOv8-based web scraper to identify and replace company logos across multiple websites.

PROJECT

Basketball Game Visualization and Performance Evaluation & Tactical Decision Support System
Feb 2025 – present

- Implemented an individual project under the supervision of the professor at King's College London
- Analyzed player shooting positions and frequencies using YOLOv8 for classification.

Classroom Roll Call Technology Based on One-shot Learning **Sep 2022 – Sep 2023**

- Researched Few-shot learning using deep learning to address face recognition challenges in a student roll call system.
- Combined data augmentation, transfer learning, and Triplet Network to train a model with one photo per student, achieving accurate face recognition.
- Proposed and verified a research plan for a fellowship from the National Science and Technology Council, Taiwan. (Approval rate of less than 5%)
- Received the Paper Acceptance of Domestic Track on Technologies and Applications of Artificial Intelligence (TAAI), Oct 2023. (Paper ID : TAAI2023-Domestic-53)

Our-Scheme (Programming Language Project) **Feb 2023 – Jul 2023**

- Built a custom C++ interpreter for processing and evaluating expressions based on formal language specifications.
- Engineered lexical analysis and syntax parsing modules to tokenize and interpret user input.

5-stage Pipelined CPU (Computer Organization Project) **Feb 2022 – Jul 2022**

- Designed and implemented a 5-stage pipelined MIPS-Lite CPU using Verilog HDL and ModelSim, supporting arithmetic, logic, memory, and branching operations, validated through waveform simulations.

SIC/SICXE Assembler (System Programming Project) **Aug 2021 – Jan 2022**

- Developed a two-pass assembler in C++ for SIC and SIC/XE machine languages, translating source code into object code with syntax validation and support for various instructions.

EXPERINENCE

4th AI&BD Innovative Competition **Taoyuan, Taiwan**
Team Leader/Finalist **Apr 2022 – Sep 2022**

- Led a cross-disciplinary team to propose and present a mental health testing software project, utilizing big data for personalized database creation and mental state detection.

Stanford University DeepLearning.AI – Coursera **Jul 2022 – Aug 2022**

- Self-studied Python and deep learning, mastering key concepts with NumPy and TensorFlow for deep neural network applications.
- Gained proficiency in using Python for image recognition tasks with deep learning models.